Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/12/2009
PI Name:	Gewirtz, Alan M. M.D.		
Project Title:	Effect of Deep Space Radiation on Human Hematopoietic Stem and Progenitor Cell Function		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gewirtz@mail.med.upenn.edu	**Fax:**	FY 215-573-2078
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-4499
Organization Name:	University of Pennsylvania		
PI Address 1:	713 BRB II/III		
PI Address 2:	421 Curie Boulevard		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104	**Congressional District:**	2
Comments:	PI passed away in 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sutherland, Betsy (Brookhaven National Laboratory)		
Grant/Contract No.:	NCC 9-58-IIH00404		
Performance Goal No.:			
Performance Goal Text:			
	(1) The original aims of this project were as follows: 1] Investigate the cellular consequences of exposing human hematopoietic stem (HSC)and progenitor (HPC)cells to an environment which simulates the radiation environment of deep space; 2] Examine the molecular consequences of exposing human hematopoietic stem cells to an environment which simulates the radiation environment of deep space; and 3] Design potential countermeasures to obviate or negate cellular and molecular damage discerned during the course of carrying out Aims 1 and 2. (2) Key Findings during last budget period: During the previous funding period, we have worked closely with Dr. Betsy Sutherland at Brookhaven National Laboratory to develop a radiation exposure model that is more relevant to the types of radiation flight crews will be exposed to during longer duration lunar and Mars missions. In this regard, we have carried out low dose, multi-dose, and mixed particule exposure of normal human hematopoietic cells to H and Fe, the main constituents of galactic cosmic rays and solar particle events. The Gewirtz Lab has tested the functional		

Task Description:	consequences of such exposure on progenitor and stem cell function in vitro and in xenograft studies employing NOD-SCID mice. The Sutherland Lab has examined effects on DNA repair pathways. Both labs have examined the effects of shielding, as well as free radical scavengers in their respective experimental systems. In brief, we have found that very low doses of either particle are tolerated but that mixed particles are much more damaging. Of note, shielding and radioprotectants appear to be able to ameliorate damage. The data supporting these statements may be found in the more detailed progress report. (3) Impact: We continued to refine our conceptual model of radiation exposure to human hematopoietic cells and to make it as realistic as possible given the constraints of the cell system we are employing (normal primary cells) and the availability of the radiation as supplied by the NSRL. We believe that the results we generated using the model available are important and support the concern that radiation exposures during EVA or SPE may be damaging.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of effective radiation countermeasures could have significant impact on earth for civilians and military personnel alike. With regard to the civilian population, it is quite conceivable that results we obtain will be relevant to patients undergoing cancer chemotherapy and radiation therapy. In this regard it is possible that our studies will provide reagents and strategies for helping to protect normal tissues from the collateral damage of anticancer treatments. It is also possible that results we generate will be relevant to radiation workers, and members of the armed forces who may be exposed to radiation during the course of carrying our their respective duties.
Task Progress:	The team of Gewirtz and Sutherland has done much to characterize the damage of various types, combinations, and doses of deep space radiations on normal human hematopoietic progenitor cells. This information will be valuable for estimating risk of catastrophic bone marrow failure and the utility of specific countermeasures for mitigating such damage. It is possible that some of this information will be useful for estimating risk of development of hematologic malignancies post-mission.
Bibliography Type:	Description: (Last Updated: 04/12/2011)
Articles in Peer-reviewed Journals	Bennett PV, Cutter NC, Sutherland BM. "Split-dose exposures versus dual ion exposure in human cell neoplastic transformation." Radiat Environ Biophys. 2007 Jun;46(2):119-23. PMID: 17256176 , Jun-2007
Articles in Peer-reviewed Journals	Paap B, Wilson DM 3rd, Sutherland BM. "Human abasic endonuclease action on multilesion abasic clusters: implications for radiation-induced biological damage." Nucleic Acids Res. 2008 May;36(8):2717-27. PMID: 18353858 , May-2008
Articles in Peer-reviewed Journals	Das B, Bennett PV, Cutter NC, Sutherland JC, Sutherland BM. "Melatonin protects human cells from clustered DNA damages, killing and acquisition of soft agar growth induced by X-rays or 970 MeV/n Fe ions." Int J Radiat Biol. 2011 Mar 14. [Epub ahead of print] PMID: 21401316 , Mar-2011
Awards	Gewirtz AM. "Elected to Association of American Physicians, August 2008." Aug-2008
Papers from Meeting Proceedings	Bennett PV, Cutter NC, Sutherland BM. "Dual ion exposure vs. split-dose exposures in human cell neoplastic transformation." 4th International Workshop on Space Radiation Research, June 2006, St. Petersburg, Russia. 4th International Workshop on Space Radiation Research, June 2006. , Jun-2006